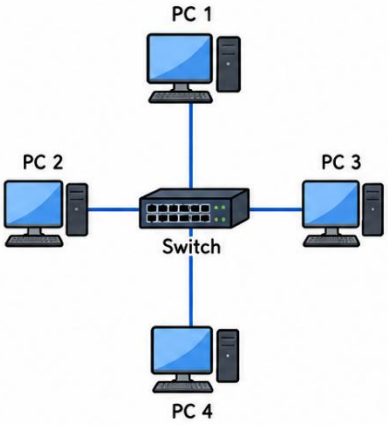
Assignment 2

Network Topologies

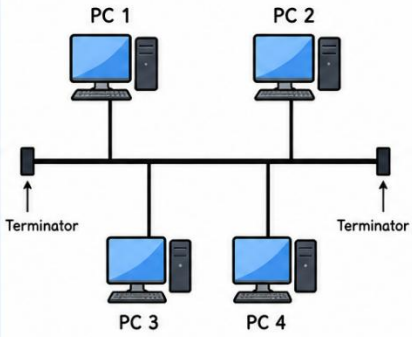
1. **Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.**

Star Topology



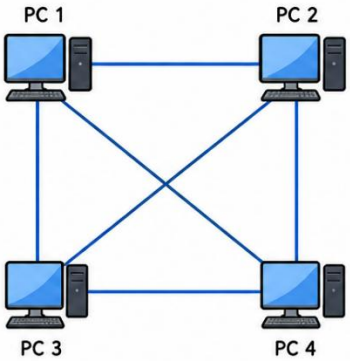
- All devices are connected to a central device (switch or hub).
- Data goes through the central device.
- Easy to manage and add new devices.
- If the central device fails, the whole network stops working.

Bus Topology



- All devices share a single cable (backbone).
- Data travels in both directions (broadcast).
- Simple and low cost.
- If the main cable fails, the whole network stops working.

Mesh Topology



- Devices are connected to many other devices.
- Provides multiple paths for data.
- Very reliable and fault tolerant.
- Expensive and more complex to set up and manage.

1. Star Topology

- **All computers are connected to a central device, usually a switch or hub.**
- **Data passes through the central device.**
- **Easy to manage, troubleshoot, and add new computers.**

2. Bus Topology

- **All computers are connected to one main cable, called a backbone.**
- **Data travels along the backbone to reach the destination.**
- **It is simple and inexpensive for small networks.**

3. Mesh Topology

- **Each computer is connected to multiple other computers.**
- **It provides multiple paths for data to travel.**
- **If one connection fails, data can use another path.**

2. **Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).**

Topology	4 Pros	4 Cons	Best Use Case	App Example
⭐ **Star**	Easy to manage, easy to expand, easy to troubleshoot, one cable failure affects one device	Central device failure, more cable, higher cost, depends on switch	Offices, schools	Instagram
🚌 **Bus**	Low cost, simple, less cable, easy for small networks	Main cable failure, slow with many devices, difficult troubleshooting, collisions	Small/old networks	Zomato
🔄 **Ring**	Organized data flow, fewer collisions, equal access, efficient	Cable failure can affect network, difficult to expand, difficult troubleshooting, less flexible	Controlled networks	Paytm
🕸️ **Mesh**	Reliable, multiple paths, fault tolerant, connection backup	Expensive, many cables, complex, difficult maintenance	Data centers, critical networks	Instagram/Paytm

3. **Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.**

- I would choose **Mesh Topology** for a multiplayer game like PUBG or Free Fire.

- It **provides multiple paths for data**, so **the game can remain reliable if one connection fails**.

- It can provide **fast communication between servers**, **reducing delays** during gameplay.

- However, it is expensive and complex to set up and maintain.

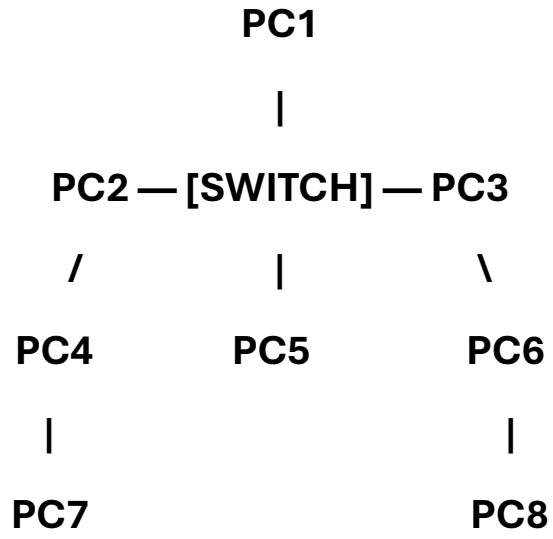
- For a multiplayer game, a **mesh server connection** could look like this:

Example:
Server A ↔ Server B
↕ ↕
Server C ↔ Server D

If **Server A** has a problem, players' data can be routed through **Server B, C, or D** instead. This makes the game more **reliable**, although setting up many server connections costs more.

4. **Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines. Hint: Consider factors like cost, ease of troubleshooting, and future expansion**

Recommended Topology: Star Topology



- **Reason**: Star topology is easy to install and troubleshoot because all computers connect to one switch. It also makes future expansion easy, as more computers can be added to the switch. Cost is reasonable for a small warehouse.